Wileam Yonatan Phan

+62 817 4950 945 · +65 8289 2635 · +1 (865) 244-5042
wileam@phan.codes
https://wyphan.github.io/ · https://linkedin.com/in/wileam-phan-389633206

Interests

- High performance computing
- Accelerated computing
- Performance analysis
- Computational material science
- Numerical algorithms
- Mathematics of arrays
- Compiler technology
- Single-board computers
- Emerging computer architectures
- Bare-metal virtualization
- Container technologies
- Continuous integration

Education

Master of Science (MS) August 2016 to December 2021

University of Tennessee, Knoxville
Department of Physics & Astronomy, College of Arts and Sciences
Knoxville, Tennessee, United States of America


MS thesis

- **Physics major** with specialization in **computational condensed-matter physics**

  Thesis: "Accelerating Dynamical Density Response Code on Summit and Its Application for Computing the Density Response Function of Vanadium Sesquioxide"

  Advisor: Prof. Adolfo G. Eguiluz (University of Tennessee, Knoxville)

  https://trace.tennessee.edu/utk_gradthes/6327/

- Recipient of Robert Birkhoff fellowship

Sarjana Sains (S.Si.) – *equivalent to* Bachelor of Science July 2009 to May 2014

Universitas Indonesia
Departemen Fisika, Fakultas Matematika dan Ilmu Pengetahuan Alam
Depok, Jawa Barat, Indonesia


S.Si. thesis

- **Physics major** with concentration in **condensed matter physics**

  Skripsi / *thesis*: "Theoretical Study on the Effects of Substrate on the Optical Conductivity of Graphene"

  Co-advisors:
  - Dr. M. Aziz Majidi (Universitas Indonesia)
  - Prof. Andrivo Rusydi (National University of Singapore)

  https://lib.ui.ac.id/detail?id=20414017&lokasi=lokal

Language Qualifications

- Human languages:
  - Bahasa Indonesia / Indonesian (native)
  - English (fluent)
  - 日本語 / Japanese (intermediate)
  - 普通话 / Mandarin Chinese (elementary)
  - 客家语 / Hakka Chinese (intermediate)
- Programming languages:
  - FORTRAN (66/77)
  - Modern Fortran (90/95/2003/2008/2018/2023)
  - C (89/99/11)
  - C++ (03/11)
  - Python
  - Bash shell scripting
  - Perl
  - MathWorks MATLAB™
  - Wolfram Mathematica™
  - HTML 4
  - JavaScript
  - Ruby
  - LaTeX
- Programming APIs:
  - MPI
  - OpenMP
  - OpenACC
  - OpenCL
  - CUDA C/C++
  - CUDA Fortran
  - HIP/ROCm
  - DPC++/SYCL

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Professional Memberships

- American Physical Society (APS)
- Association for Computing Machinery (ACM)
- ACM Special Interest Group on High Performance Computing (SIGHPC)

Work Experience

Software Developer (*contract*) September 2024 to August 2025

Advanced Micro Devices
Markham, Ontario, Canada (*remote*)

- Member of development team for ROCm Systems Profiler (`rocprofiler-systems`, formerly Omnitrace)
- Provided ROCm System Profiler support for HPC customers
- Participated in GPU hackathons (DiRAC, EPCC) as part of support vendor team
- Contributed to the testing and development of ROCm System Profiler

Software Developer September 2024 to August 2025

DXC Luxoft
Singapore

- Client: Advanced Micro Devices (AMD)

Gap due to immigration process in Singapore July 2024 to September 2024

Research Software Engineer June 2022 to July 2024

Rice University
Department of Computer Science, George R. Brown School of Engineering
Houston, Texas, United States of America (*remote*)

- Member of development team for HPCToolkit profiling tools, part of Exascale Computing Project (ECP)
- Maintained HPCToolkit deployments at DOE open-science leadership computing facilities
- Led application engagement activities to collaborate with ECP application codes
- Collected feedback on HPCToolkit usage from ECP application teams
- Participated in GPU hackathons (ORNL, NERSC) as part of support vendor team
- Prepared and delivered user trainings and workshops
- Served as ECP project coordinator for the HPCToolkit project
- Contributed to the research and development of HPCToolkit for GPU-accelerated applications

Research Software Engineer (*part-time, contract*) March 2022 to May 2022

Sourcery Institute
Oakland, California, United States of America (*remote*)

- Isolated Fortran 2018/2023 implementation bugs in GCC GFortran compiler and wrote reproducer codes
- Wrote Fortran 2018 standard compliance tests

Research Assistant March 2021 to June 2022

University at Albany, State University of New York
Department of Computer Science, College of Nanotechnology, Science, and Engineering
Albany, New York, United States of America (*remote*)

- Assisted the GPU porting process for codes based on Mathematics of Arrays

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Work Experience (*continued*)

Scientific Computing Software Engineer / CSE-2 July 2021 to October 2021

Lawrence Berkeley National Laboratory
Center for Computational Sciences and Engineering, Computational Research Division
Berkeley, California, United States of America (*remote*)

- Member of development team for Adaptive Mesh Refinement for Exascale (AMReX) project, part of ECP
- Ported several code components from Fortran to C++ with GPU support
- Developed unit tests, which were executed on Gigan (CCSE) and Spock (OLCF)
- Contributed to user documentation

Graduate Research Assistant August 2020 to July 2021

University of Tennessee, Knoxville
Department of Physics & Astronomy, College of Arts and Sciences
Knoxville, Tennessee, United States of America

- Developed and ported the Eguiluz research group "Exciting-Plus" density response code for execution on graphic processing units (GPUs), using OpenACC and GPU libraries (MAGMA), targeting the Summit supercomputer at Oak Ridge Leadership Computing Facility (OLCF)
- Participated as member of team EECM in the 2020 OLCF GPU Hackathon
  https://www.olcf.ornl.gov/2020-olcf-gpu-hackathon/
- Performed calculations with the ported code on Summit (OLCF) and Cori-GPU (NERSC)
- Achieved up to 12× speed-up over original CPU-only version

Academic summer break at University of Tennessee, Knoxville May 2020 to August 2020

Graduate Teaching Assistant August 2016 to May 2020

University of Tennessee, Knoxville
Department of Physics & Astronomy, College of Arts and Sciences
Knoxville, Tennessee, United States of America

- Taught physics laboratory sessions (both traditional and hybrid studio methods) for the following courses:
  - PHYS 221 Elements of Physics I (Fall 2017, Fall 2018, Spring 2019)
  - PHYS 222 Elements of Physics II (Spring 2017, Spring 2018, Fall 2019)
  - PHYS 231 Fundamentals of Physics I: Electricity and Magnetism (Fall 2016, Spring 2018)
- Graded for the following course:
  - PHYS 514 Problems in Theoretical Physics II (Spring 2020)

Gap due to preparations for admission into University of Tennessee, Knoxville December 2015 to August 2016

Teaching Assistant January 2011 to December 2015

Universitas Indonesia
Departemen Fisika, Fakultas Matematika dan Ilmu Pengetahuan Alam
Depok, Jawa Barat, Indonesia

- Held tutorials, proctored exams, and graded homework & exams for the following courses:
  - FSK 20236 Electromagnetic Fields 1 (January 2011 to June 2013)
  - SCFI 603611 Solid State Physics 1 (August 2014 to December 2015)
  - SCFI 604021 Computational Physics 2 (August 2015 to December 2015)

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Publications

ORCID: 0000-0001-5621-5949

- L Adhianto, J Anderson, RM Barnett, D Grbic, V Indic, M Krentel, Y Liu, S Milaković, W Phan, J Mellor-Crummey
  "Refining HPCToolkit for Application Performance Analysis at Exascale"
  The International Journal of High Performance Computing Applications **38** (6), 612–632 (2024)
  doi: 10.1177/10943420241277839
- LM Mullin and WY Phan
  "A Transformational Approach to Scientific Software:
  the Mathematics of Arrays (MoA) Fast Fourier Transform (FFT) with OpenACC"
  Invited talk at OpenACC Summit 2021
  https://www.openacc.org/events/openacc-summit-2021
- MA Majidi, R Kusumaatmadja, AD Fauzi, WY Phan, A Taufik, R Saleh, and A Rusydi
  "Theoretical Exploration of Optical Response of Fe_3O_4-reduced Graphene Oxide Nanoparticle System within Dynamical Mean-Field Theory"
  IOP Conference Series: Materials Science & Engineering **188**, 012055 (2017)
  doi: 10.1088/1757-899x/188/1/012055
- MA Majidi, WY Phan, and A Rusydi
  "Investigation of the Effects of the Graphene-Substrate Hybridization on the Optical Conductivity of Graphene"
  AIP Conference Proceedings **1729**, 020016 (2016)
  doi: 10.1063/1.4946919
- MA Majidi, MA Naradipa, WY Phan, A Syahroni, and A Rusydi
  "Development of Tight-binding Based GW Algorithm and Its Computational Implementation for Graphene"
  AIP Conference Proceedings **1729**, 020013 (2016)
  doi: 10.1063/1.4946916

Trainings Given

- John Mellor-Crummey and Wileam Phan
  "HPCToolkit: Emerging Performance Tools for Exascale Computing" (February 6, 2023)
  Tutorial at ECP Annual Meeting 2022 Project Tutorial Days
  https://www.exascaleproject.org/event/ecp-tutorial-days/
- John Mellor-Crummey and Wileam Phan
  "HPCToolkit Overview" (December 14, 2022)
  Presentation at Oak Ridge Leadership Computing Facility (OLCF) User Conference Call
  https://www.olcf.ornl.gov/calendar/userconcall-dec2022/
- Wileam Phan
  "Towards Exascale Computing: Recent Trends in High-Performance Computing (HPC)" (February 9, 2022)
  Seminar at Departemen Fisika, Universitas Indonesia
  https://wyphan.github.io/logfile/2022/02/09/fisika-ui-talk.html

Hackathons Supported

- EPCC AMD MI300 Series Hackathon (May 27-29, 2025)
  Edinburgh Parallel Computing Centre – remote participation
  https://www.epcc.ed.ac.uk/whats-happening/articles/amd-mi300-series-hackathon
- DiRAC Hackathon (April 2-4, 2025)
  DiRAC High Performance Computing Facility – remote participation

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Hackathons Supported (*continued*)

- June 2023 Frontier Hackathon (June 28-30, 2023)
  Oak Ridge Leadership Computing Facility (OLCF) – remote participation
  https://www.olcf.ornl.gov/frontier-hackathon-june-2023/
- Aurora COE Workshop 5 (June 13-15, 2023)
  Argonne Leadership Computing Facility (ALCF) – remote participation
  https://events.cels.anl.gov/event/404/
- NERSC December 2022 GPU Hackathon (November 30 & December 6-8, 2022)
  National Energy Research Scientific Computing Center (NERSC) – remote participation
  https://sites.google.com/lbl.gov/december2022gpuhackathon/home
- Crusher Hackathon 4 (September 12-14, 2022)
  Oak Ridge Leadership Computing Facility (OLCF) – remote participation
  https://www.olcf.ornl.gov/crusher-hackathon-4/

Trainings and Workshops Attended

- Writing Clean Scientific Software (July 12, 2023)
  Exascale Computing Project (ECP) – remote participation
  https://www.exascaleproject.org/event/cleanssw/
- FUN Training July 2023: Modern Fortran Basics (July 10-11, 2023)
  National Energy Research Scientific Computing Center (NERSC) – remote participation
  https://www.nersc.gov/users/training/events/fun-training-july-2023-modern-fortran-basics/
- Lessons Learned Developing Performance-Portable QMCPACK (May 10, 2023)
  Exascale Computing Project (ECP) – remote participation
  https://www.exascaleproject.org/event/qmcpack/
- Python - Distribution & Packaging Roundtable (May 9, 2023)
  OpenTeams – remote participation
  https://hopin.com/events/python-distribution-packaging-roundtable-20230509/registration
- Architecting for Accelerators: How to Use Intel's New, Built-in AI Acceleration Engines (March 29, 2023)
  Intel – remote participation
- Ookami User Group Meeting 2023 (March 23, 2023)
  Institute for Advanced Computational Science, Stony Brook University – remote participation
  https://www.stonybrook.edu/commcms/ookami/user_meeting/UGM2023.php
- Frontier Training Workshop (February 15-17, 2023)
  Oak Ridge Leadership Computing Facility (OLCF) – remote participation
  https://www.olcf.ornl.gov/calendar/frontier-training-workshop-february-2023/
- Using HIP and GPU Libraries with OpenMP Webinar (December 14, 2022)
  Oak Ridge Leadership Computing Facility (OLCF) – remote participation
  https://www.olcf.ornl.gov/calendar/preparing-for-frontier-openmp-part3/
- Crusher User Experience Talks (December 9, 2022)
  Oak Ridge Leadership Computing Facility (OLCF) – remote participation
  https://www.olcf.ornl.gov/calendar/crusher-user-experience-talks/
- Black Lives Count: Toward Accountability in Efforts to Diversify Computing (October 25, 2022)
  Exascale Computing Project (ECP) – remote participation
  https://www.exascaleproject.org/event/blc/
- Hierarchical Roofline Profiling on AMD GPUs Webinar (October 11, 2022)
  Oak Ridge Leadership Computing Facility (OLCF) – remote participation
- Solve the Mystery of Performance Profiling Webinar (September 28, 2022)
  Intel – remote participation

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Trainings and Workshops Attended (*continued*)

- SYCL Essentials – oneDPL Coding Webinar (September 20, 2022)
  Intel – remote participation
- Software Packaging Webinar (September 7, 2022)
  Exascale Computing Project (ECP) – remote participation
  https://www.exascaleproject.org/event/softwarepackaging/
- Focus on Ally Skills Webinar (August 30, 2022)
  Exascale Computing Project (ECP) – remote participation
  https://www.exascaleproject.org/event/focus-on-ally-skills/
- What Can Be Learned from Applying Team of Teams Principles to the ECP projects PETSc, Trillinos, xSDK, and E4S? (August 25, 2022)
  Exascale Computing Project (ECP) – remote participation
  https://www.exascaleproject.org/event/team-of-teams1/
- Understanding GPU Register Pressure Webinar (August 22, 2022)
  Oak Ridge Leadership Computing Facility (OLCF) – remote participation
- SYCL Essentials – oneAPI Advanced Concepts Webinar (August 18, 2022)
  Intel – remote participation
- SYCL Essentials – oneAPI Introduction Webinar (July 26, 2022)
  Intel – remote participation
- Introduction to Slingshot (July 13, 2022)
  Exascale Computing Project (ECP) – remote participation
- Introduction to the Frontier Supercomputer Webinar (July 12, 2022)
  Oak Ridge Leadership Computing Facility (OLCF) – remote participation
  https://www.olcf.ornl.gov/calendar/introduction-to-the-frontier-supercomputer/
- Profiling Deep Learning Applications with NVIDIA Nsight Webinar (June 30, 2022)
  Argonne Leadership Computing Facility (ALCF) – remote participation
  https://www.alcf.anl.gov/events/profiling-deep-learning-applications-nvidia-nsight
- 14th Scalable Tools Workshop (June 19-23, 2022)
  Lake Tahoe, California, United States of America
  https://dyninst.github.io/scalable_tools_workshop/petascale2022/
- Crash Course in Supercomputing Workshop (June 14, 2022)
  National Energy Research Scientific Computing Center (NERSC) – remote participation
  https://www.nersc.gov/users/training/events/crash-course-in-supercomputing-jun2022/
- Graviton2 Arm vs Intel x86: Comparing the Performance of Graviton-based Systems vs x86 Hardware for Real-time Streaming Data Applications Webinar (June 8, 2022)
  Linux Foundation – remote participation
  https://www.linuxfoundation.org/webinars/graviton2-arm-vs-intel-x86/
- How to be a Great Mentor Webinar (May 24, 2022)
  Exascale Computing Project (ECP) – remote participation
  https://www.exascaleproject.org/event/mentor/
- Coding for GPUs Using Standard Fortran Webinar (May 13, 2022)
  Oak Ridge Leadership Computing Facility (OLCF) – remote participation
  https://www.olcf.ornl.gov/calendar/coding-for-gpus-using-standard-fortran/

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Trainings and Workshops Attended (*continued*)

- Coding for GPUs Using Standard C++ Webinar (April 7, 2022)
  Oak Ridge Leadership Computing Facility (OLCF) – remote participation
  https://www.olcf.ornl.gov/calendar/coding-for-gpus-using-standard-c/
- 8th BerkeleyGW Tutorial Workshop (January 10-12, 2022)
  Lawrence Berkeley National Laboratory (LBL) – remote participation
  https://workshop.berkeleygw.org/
- Using Perlmutter Training Workshop (January 5-7, 2022)
  National Energy Research Scientific Computing Center (NERSC) – remote participation
  https://www.nersc.gov/users/training/events/using-perlmutter-training-jan2022/
- ECP CMake Training Workshop (August 23-26, 2021)
  National Energy Research Scientific Computing Center (NERSC) – remote participation
  https://www.nersc.gov/users/training/events/ecp-cmake-training-aug-2021/
- Introduction to CI at NERSC Training (July 7, 2021)
  National Energy Research Scientific Computing Center (NERSC) – remote participation
  https://www.nersc.gov/users/training/events/introduction-to-ci-at-nersc-july-7-2021/
- Perlmutter Introduction Training (June 2, 2021)
  National Energy Research Scientific Computing Center (NERSC) – remote participation
  https://www.nersc.gov/users/training/events/perlmutter-introduction-june-2021/
- HIP Training Workshop (May 24-26, 2021)
  Oak Ridge Leadership Computing Facility (OLCF) – remote participation
  https://www.olcf.ornl.gov/calendar/2021hip
- Automated Fortran-C++ Bindings for Large Scale Scientific Applications (May 12, 2021)
  Exascale Computing Project – remote participation
  https://www.exascaleproject.org/event/fortran-cpp-bindings/
- AMD EPYC Advanced User Training on Expanse (April 21, 2021)
  San Diego Supercomputing Center (*SDSC*) – remote participation through NSF XSEDE
  https://www.xsede.org/web/xup/course-calendar/-/training-user/class/2311
- Using HPCToolkit to Measure and Analyze the Performance of GPU-accelerated Applications Tutorial (March 29 & April 2, 2021)
  National Energy Research Scientific Computing Center (NERSC) – remote participation
  https://www.nersc.gov/users/training/events/hpctoolkit-for-gpu-tutorial-mar-apr-2021
- Introduction to Ookami Webinar (March 3, 2021)
  Institute for Advanced Computational Science, Stony Brook University – remote participation
- Good Practices for Research Software Documentation Webinar (February 10, 2021)
  Exascale Computing Project – remote participation
  https://ideas-productivity.org/events/hpc-best-practices-webinars/#webinar049
- Extreme-scale Scientific Software Stack (E4S) Webinar (January 13, 2021)
  Exascale Computing Project – remote participation
  https://www.exascaleproject.org/event/e4s-210113/

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- Totalview Tutorial (December 9, 2020)
  National Energy Research Scientific Computing Center (NERSC) – remote participation
  https://www.nersc.gov/users/training/events/totalview-tutorial-december-9-2020/

Trainings and Workshops Attended (*continued*)

- 2020 OLCF GPU Hackathon (October 19 & 26-28, 2020)
  Oak Ridge Leadership Computing Facility (OLCF) – remote participation
  https://www.olcf.ornl.gov/2020-olcf-gpu-hackathon/
- CUDA Training Series (January - September 2020; July - October 2021)
  Oak Ridge Leadership Computing Facility (OLCF) – remote and on-site participation
  https://www.olcf.ornl.gov/cuda-training-series/
- Testing and Code Review Practices in Research Software Development Webinar (September 9, 2020)
  Exascale Computing Project – remote participation
  https://www.exascaleproject.org/event/testing-and-code-review/
- TAU Performance Analysis Training (July 28, 2020)
  Oak Ridge Leadership Computing Facility (OLCF) – remote participation
  https://www.olcf.ornl.gov/calendar/tau-performance-analysis-training/
- Arm Debugging and Profiling Tools Tutorial (July 16, 2020)
  National Energy Research Scientific Computing Center (NERSC) – remote participation
  https://www.nersc.gov/users/training/events/arm-debugging-and-profiling-tools-tutorial-june-25-2020/
- OpenACC Training Series (April - June 2020, once a month)
  Oak Ridge Leadership Computing Facility (OLCF) – remote participation
  https://www.olcf.ornl.gov/openacc-training-series/
- NVIDIA Profiling Tools – Nsight Compute Training (March 10, 2020)
  Oak Ridge Leadership Computing Facility (OLCF) – remote participation
  https://www.olcf.ornl.gov/calendar/nvidia-profiling-tools-nsight-compute/
- NVIDIA Profiling Tools – Nsight Systems Training (March 9, 2020)
  Oak Ridge Leadership Computing Facility (OLCF) – remote participation
  https://www.olcf.ornl.gov/calendar/nvidia-profiling-tools-nsight-systems/
- Programming for Advanced Architectures on Stampede2 Training (October 31, 2018)
  Texas Advanced Computing Center (TACC) – remote participation through NSF XSEDE
  https://portal.xsede.org/course-calendar/-/training-user/class/880
- Workshop Klaster Komputer 2013 (December 11-13, 2013)
  Lembaga Ilmu Pengetahuan Indonesia (LIPI), Bandung, Jawa Barat, Indonesia
  http://situs.opi.lipi.go.id/wkk2013/